Module 42

Indexing and Hashing/2: Indexing/2

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Module Recap

- Appreciated the reasons for indexing database tables

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- Understood the ordered indexes

Module Objectives

- To recap Balanced Binary Search Trees as options for optimal in-memory search data structures

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- To understand the issues relating to external search data structures for persistent data
- To study 2-3-4 Tree as a precursor to B/B+-Tree for an efficient external data structure for database and index tables

Module Outline

- Balanced Binary Search Trees

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- 2-3-4 Tree

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- `01 05 08 10 12 15 20 22 25 28 30 36 38 40 45 48 50 END`
- ▷ **Binary Search Tree** - recursively on left / right

Search Data Structures (2)

- **Worst case time** (n data items in the data structure):

Data Structure	Search	Insert	Delete	Remarks
Unordered Array	$\mathcal{O}(n)$	$\mathcal{O}(1)$	$\mathcal{O}(1)$	Time after location is ascertained by Search
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Unordered List	$\mathcal{O}(n)$	$\mathcal{O}(1)$	$\mathcal{O}(n)$	Time after location is ascertained by Search
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  - - A BST where recently accessed elements are quick to access again

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  - Has complex rotation and / or similar operations
  - Do not scale for external data structures

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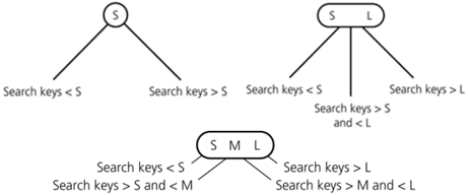

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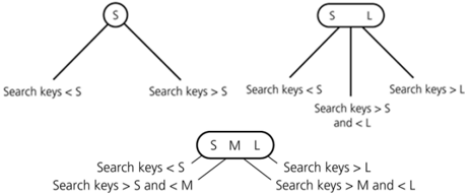

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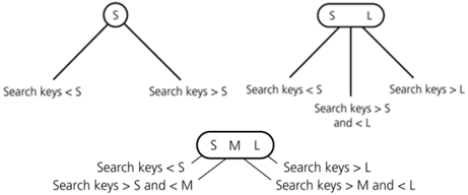

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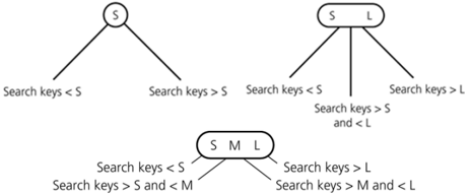

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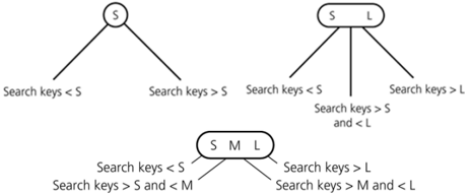

2-3-4 Trees: Search

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  - Simple and natural extension of search in BST

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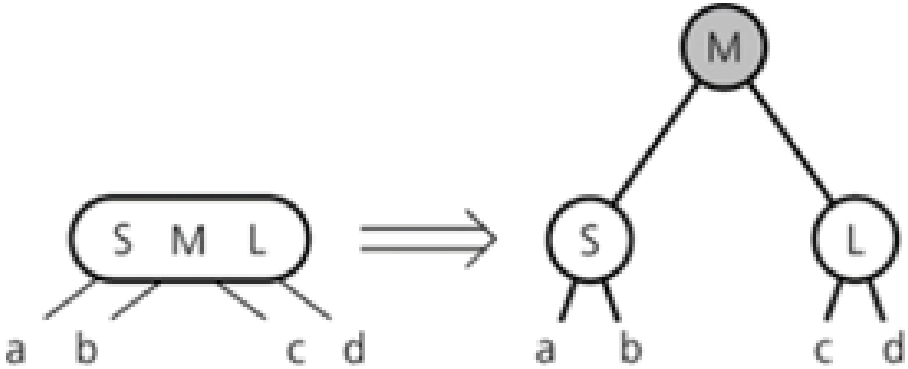
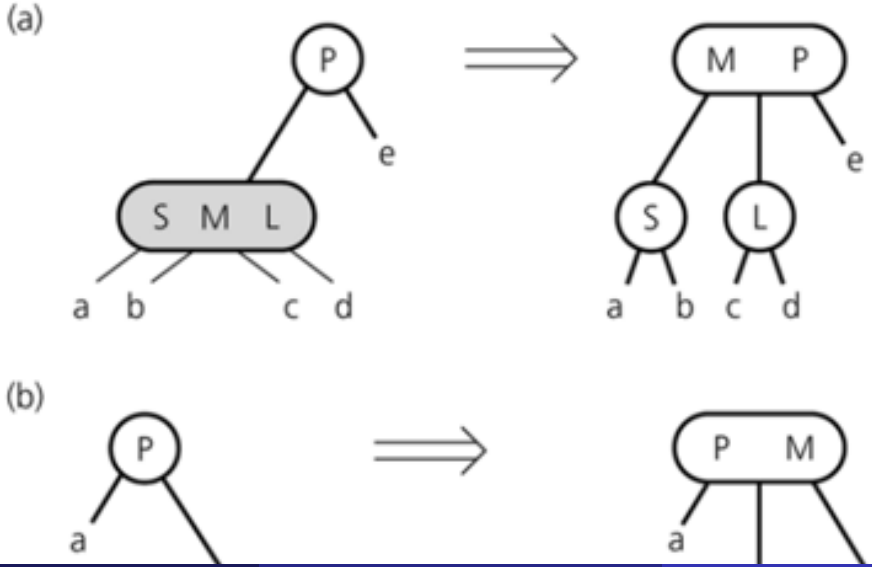
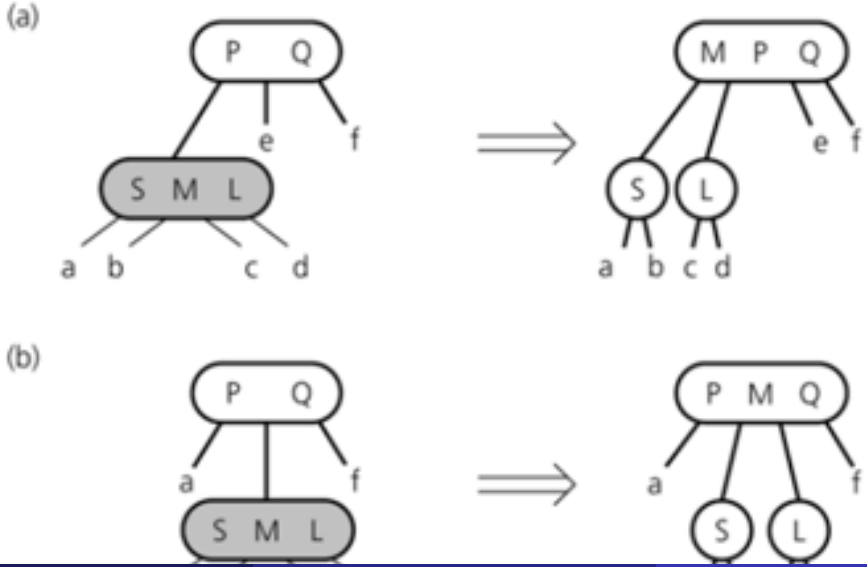
2-3-4 Trees: Insert - Splitting at Root



2-3-4 Trees: Insert - Splitting with 2 Node parent



2-3-4 Trees: Insert - Splitting with 3 Node parent



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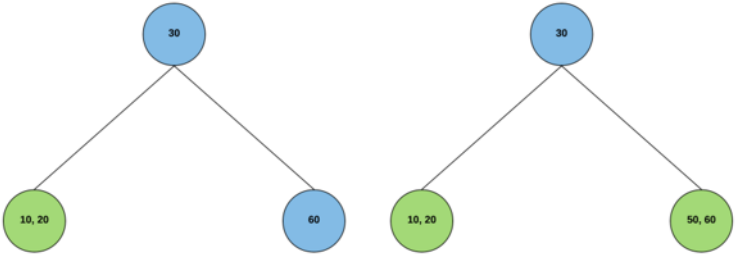
- Here we are following **early strategy**

2-3-4 Trees: Insert: Example

Example

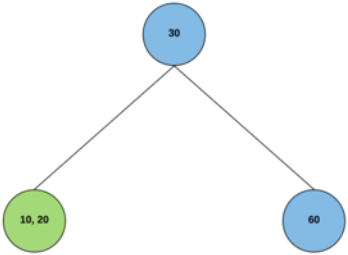
- Insert: 10, 30, 60, 20

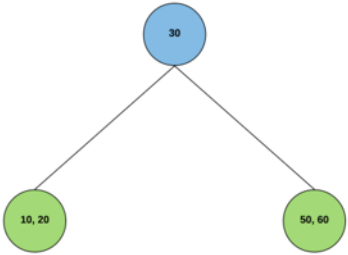


2-3-4 Trees: Insert: Example

Example

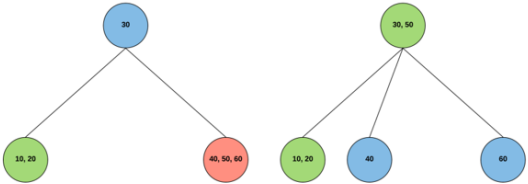
- Insert: 10, 30, 60, 20
- Insert: 50





2-3-4 Trees: Insert: Example (Continued)

- 10, 30, 60, 20, 50, 40
- Split for 70



2-3-4 Trees: Insert: Example (Continued)

Example

- Insert: 40, 70

2-3-4 Trees: Insert: Example (Continued)

Example

- Insert: 40, 70
- Insert: 80

2-3-4 Trees: Insert: Example (Continued)

Example

- Insert: 15

2-3-4 Trees: Insert: Example (Continued)

Example

- Insert: 15
- Split for 90

2-3-4 Trees: Insert: Example (Continued)

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- Insert: 90

2-3-4 Trees: Insert: Example (Continued)

Example

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- Split for 100

2-3-4 Trees: Insert: Example (Continued)

Example

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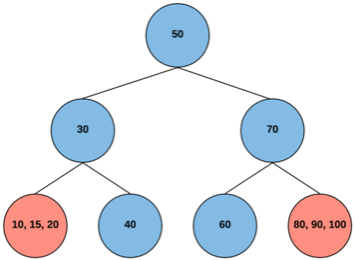

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- All leaves are at the same depth (the bottom level): Height, $h \sim \mathcal{O}(\log n)$

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- Uses variety of node types - need to destruct and construct multiple nodes for converting a 2 Node to 3 Node, a 3 Node to 4 Node, for splitting etc.

2-3-4 Trees (continued)

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  - **B-Tree**
  - 2-3-4 Tree is a B-Tree where $n = 4$

Module Summary

- Recapitulated the notions of Balanced Binary Search Trees as options for optimal in-memory search data structures

Slides used in this presentation are borrowed from `http://db-book.com/` *with kind permission of the authors.*
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Module Summary

- Recapitulated the notions of Balanced Binary Search Trees as options for optimal in-memory search data structures
- Understood the issues relating to external data structures for persistent data
- Explored 2-3-4 Tree in depth as a precursor to B/B+-Tree for an efficient external data structure for database and index tables

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